DSA Algorithms Time Complexity Cheat Sheet

'*' Union-Find with path compression and union by rank has inverse Ackermann complexity $\alpha(n)$, effectively constant for practical inputs.

Algorithm / Operation	Best	Average	Worst
Linear Search	O(1)	O(n)	O(n)
Binary Search	O(1)	O(log n)	O(log n)
Bubble Sort	O(n)	$O(n^2)$	$O(n^2)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Insertion Sort	O(n)	$O(n^2)$	$O(n^2)$
Merge Sort	O(n log n)	O(n log n)	O(n log n)
Quick Sort	O(n log n)	O(n log n)	$O(n^2)$
Heap Sort	O(n log n)	O(n log n)	O(n log n)
Counting Sort	O(n+k)	O(n+k)	O(n+k)
Radix Sort	O(d(n+k))	O(d(n+k))	O(d(n+k))
Bucket Sort	O(n+k)	O(n+k)	$O(n^2)$
DFS	O(V+E)	O(V+E)	O(V+E)
BFS	O(V+E)	O(V+E)	O(V+E)
Dijkstra (Heap)	O(E log V)	O(E log V)	O(E log V)
Bellman-Ford	O(VE)	O(VE)	O(VE)
Floyd-Warshall	$O(V^3)$	$O(V^3)$	$O(V^3)$
Kruskal	O(E log E)	O(E log E)	O(E log E)
Prim (Heap)	O(E log V)	O(E log V)	O(E log V)
Topological Sort	O(V+E)	O(V+E)	O(V+E)
Union-Find (find/union)	O(1)*	$O(\alpha(n))$	$O(\alpha(n))$
Trie Search	O(L)	O(L)	O(L)
Trie Insert	O(L)	O(L)	O(L)
KMP Search	O(n+m)	O(n+m)	O(n+m)
Rabin-Karp	O(n+m)	O(n+m)	O(nm)
Segment Tree Query	O(log n)	O(log n)	O(log n)
Segment Tree Update	O(log n)	O(log n)	O(log n)
Fenwick Tree Query	O(log n)	O(log n)	O(log n)
Fenwick Tree Update	O(log n)	O(log n)	O(log n)
Kadane's Algorithm	O(n)	O(n)	O(n)
Sliding Window	O(n)	O(n)	O(n)
Two Pointers	O(n)	O(n)	O(n)

Prefix Sum Query	O(1)	O(1)	O(1)
Prefix Sum Build	O(n)	O(n)	O(n)
Binary Exponentiation	O(log n)	O(log n)	O(log n)
Euclidean GCD	O(log min(a,b))	O(log min(a,b))	O(log min(a,b))